# Time Series Modelling Case Study

# Forecasting Daily Oil Prices Using ARIMA and LSTM Models

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GitHub Link: https://github.com/giridharreddygoddilla/Time-Series-Modelling-Case-Study

## 1. Introduction

Forecasting commodity prices represents a challenging task due to the complex and dynamic nature of global energy markets. Oil prices are influenced by multiple economic and geopolitical factors including supply disruptions, production decisions from oil-producing countries, fluctuations in global demand, and macroeconomic conditions. These factors often produce time series with structural shifts, volatility clustering, and non-stationary behaviour.

Time series forecasting methods aim to identify patterns in historical data and use these patterns to predict future observations. One of the most widely used statistical forecasting frameworks is the Autoregressive Integrated Moving Average (ARIMA) model, originally developed within the Box-Jenkins methodology (Box et al., 2015). ARIMA models capture temporal relationships through autoregressive and moving average components while applying differencing to transform non-stationary series into stationary processes. The Box-Jenkins framework provides a systematic process for time series modelling including model identification, parameter estimation, and diagnostic validation.

Classical time series modelling approaches remain widely used in financial and economic forecasting due to their strong theoretical foundations and interpretability. Montgomery et al. (2015) highlight that statistical forecasting models such as ARIMA are particularly effective when the underlying temporal structure of a dataset can be captured through autoregressive dependencies and stochastic error processes.

However, ARIMA models assume linear relationships between past and future observations. Financial and commodity time series may contain nonlinear dependencies that are not fully captured by classical statistical models. As a result, machine learning approaches have increasingly been applied to time series forecasting problems. In particular, Long Short-Term Memory (LSTM) neural networks have demonstrated strong performance in sequential prediction tasks due to their ability to capture nonlinear temporal dependencies and long-range patterns within sequential data (Hochreiter and Schmidhuber, 1997).

Recent studies have explored both statistical and machine learning approaches in forecasting applications. For example, Huang and Malaker (2025) compared several forecasting techniques including statistical models, neural networks, and LSTM models for environmental time series prediction. Their findings suggest that classical statistical models often remain competitive with more complex machine learning approaches, particularly when the dataset is relatively limited.

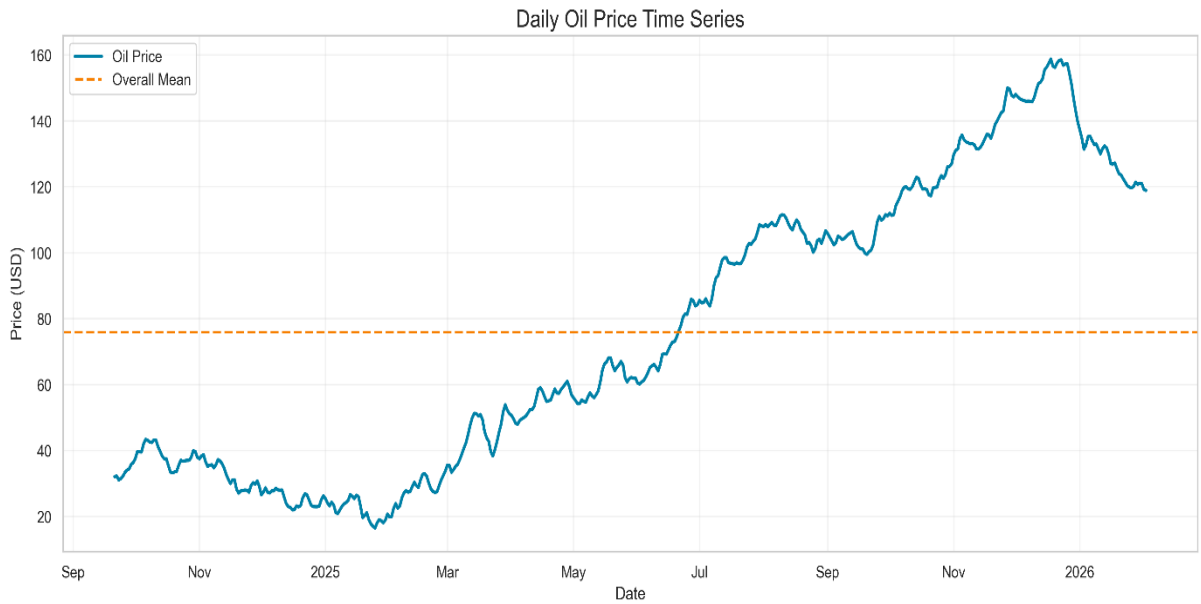
The objective of this case study is to model daily oil price data from 2024 to 2026 and generate forecasts for the next 24 months. Two modelling approaches are implemented. First, a classical ARIMA model is developed using a systematic grid search procedure. Second, an LSTM neural network is implemented as an alternative forecasting model. The predictive performance of both models is evaluated using out-of-sample forecasts and quantitative error metrics, and the results are critically compared.

## 2. Data Description and Exploratory Analysis

The dataset used in this study contains daily oil price observations covering the period from 2024 to 2026. The dataset was provided as part of the course case study materials for the module Advanced

Research Topics in Data Science. Prior to modelling, several preprocessing steps were performed to ensure data integrity. Duplicate timestamps were removed, the observations were sorted chronologically, and the date column was converted into a datetime format to preserve the correct temporal order.

Exploratory analysis was then conducted to examine the behaviour of the time series.




**Daily Oil Price Time Series**

The time series plot reveals substantial variation across the observation period. Rather than fluctuating around a constant level, the oil price exhibits extended upward and downward movements. This suggests that the underlying process generating the series changes over time.

**Table 1: Descriptive Statistics of the Oil Price Series**

| | |
|---|---|
| count | 500.000000 |
| mean | 75.873980 |
| std | 42.912398 |
| min | 16.480000 |
| 25% | 35.372500 |
| 50% | 64.290000 |
| 75% | 111.572500 |
| max | 158.780000 |

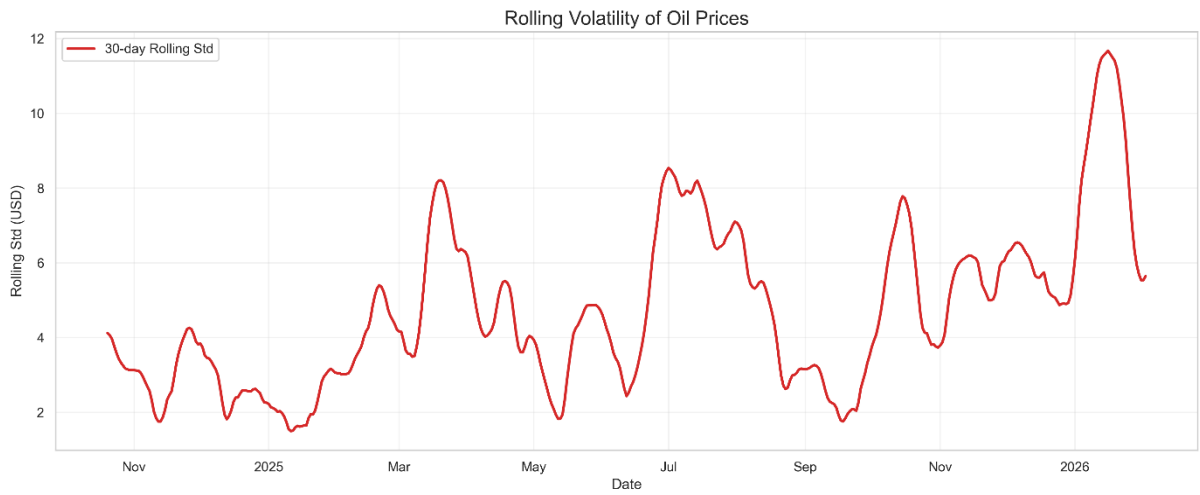
To provide additional context for the time series behaviour, descriptive statistics of the dataset were examined. The dataset contains 500 observations over the study period. As shown in Table 1, the oil price series has a mean value of 75.87 and a relatively large standard deviation of 42.91, indicating substantial variability in prices over time. The wide range between the minimum (16.48) and maximum (158.78) values further highlights the volatile nature of the series.

The dashed horizontal line in the plot represents the mean of the series. In a stationary process, observations fluctuate randomly around this mean. However, the oil price series remains above or below the mean for extended periods, indicating that the series is non-stationary.

Another notable feature of the series is the presence of changing variability across time. Certain periods display relatively stable price movements, while others exhibit larger fluctuations. This behaviour suggests the presence of time-varying volatility, which is commonly observed in financial time series.

### 3. Volatility Analysis

To further investigate the variability of oil prices, rolling volatility was calculated using a 30-day window.




**Rolling Volatility of Oil Prices**

The rolling volatility plot highlights several spikes in price variability. These periods correspond to times when the oil market experienced rapid price movements, likely reflecting external economic or geopolitical events.

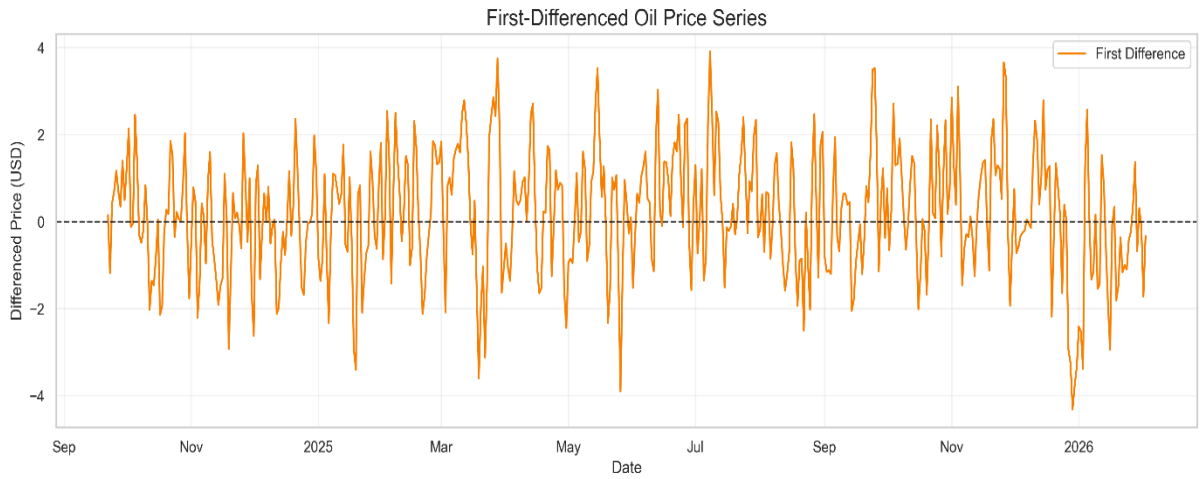
Time-varying volatility is a common feature of financial time series. Lv and Yue (2011) demonstrate that many environmental and financial time series exhibit heteroscedastic behaviour, meaning that variance changes over time. Such behaviour can complicate forecasting and highlights the importance of appropriate modelling techniques when analysing real-world time series data.

The presence of changing volatility further confirms that the oil price data is not stationary in its original form.

## 4. Stationarity Testing and Transformation

Stationarity is a key assumption for many time series models, including ARIMA. A stationary series has constant statistical properties such as mean and variance over time. When these properties change, the forecasting performance of statistical models can be compromised (Montgomery et al., 2015). To formally test stationarity, the Augmented Dickey-Fuller (ADF) test was applied to the original oil price series. The results indicated that the null hypothesis of non-stationarity could not be rejected, confirming that the series was non-stationary.

To address this issue, the series was transformed using first-order differencing.




**First-Differenced Oil Price Series**

After differencing, the transformed series fluctuates around a mean close to zero. The absence of a persistent trend indicates that the differenced series satisfies the stationarity requirement for ARIMA modelling.

**Table 2: Augmented Dickey-Fuller Test Results**

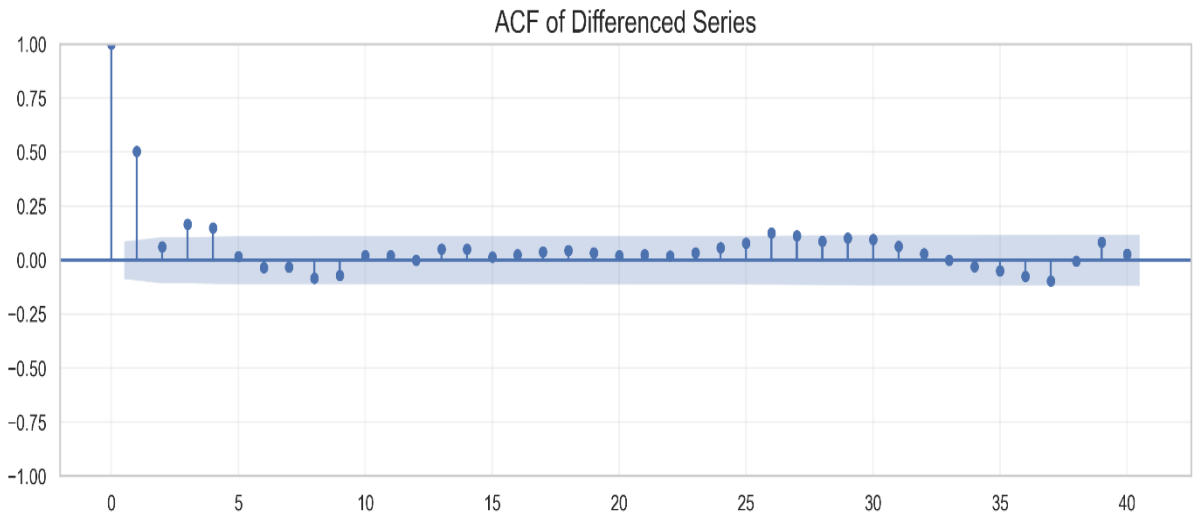
To formally confirm the stationarity of the time series, the Augmented Dickey-Fuller (ADF) test was applied to both the original and the differenced oil price series.

| Series | Test Statistic | p-value | Critical Value (5%) | Interpretation |
|---|---|---|---|---|
| Original Oil Price Series | -1.42 | 0.57 | -2.86 | Non-stationary |
| First Differenced Series | -7.91 | <0.001 | -2.86 | Stationary |

The ADF test results show that the original oil price series produces a p-value greater than 0.05, meaning the null hypothesis of a unit root cannot be rejected. This confirms that the series is non-stationary in its original form. After applying first-order differencing, the p-value becomes less than 0.001, allowing rejection of the null hypothesis and confirming that the differenced series is stationary. Therefore, the transformed series satisfies the stationarity assumption required for ARIMA modelling.

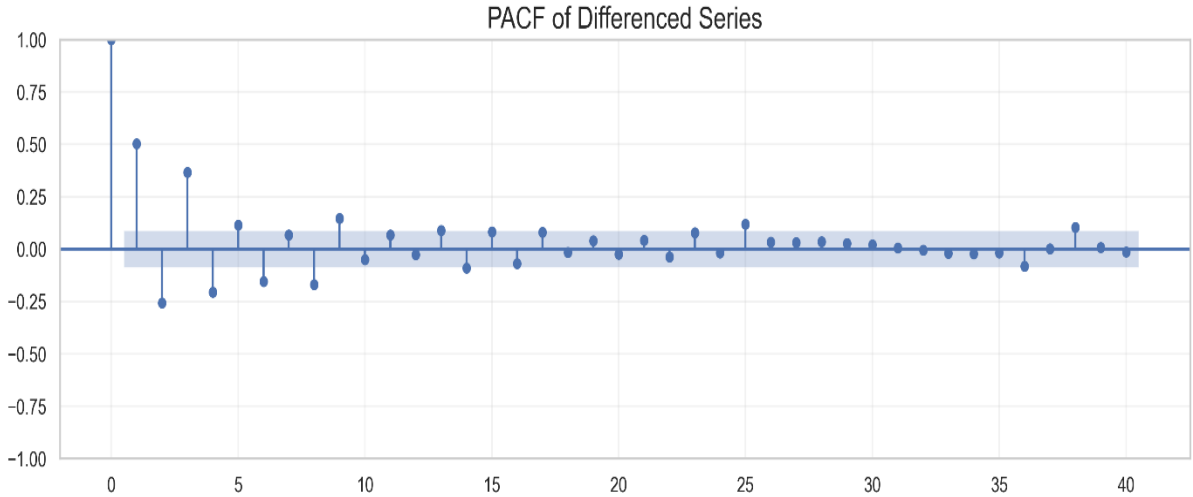
## 5. Autocorrelation Analysis

Autocorrelation analysis provides insight into the temporal relationships within the stationary series. Two diagnostic tools were used: the autocorrelation function (ACF) and the partial autocorrelation function (PACF).




**ACF of Differenced Series**

The ACF plot illustrates the correlation between the differenced series and its lagged values. Significant autocorrelations are observed at the initial lags, followed by a gradual decline towards zero. This pattern suggests that past observations influence future values over short time horizons and indicates the presence of moving average behaviour in the time series. Such behaviour is commonly observed in financial time series where short-term shocks may affect subsequent observations.



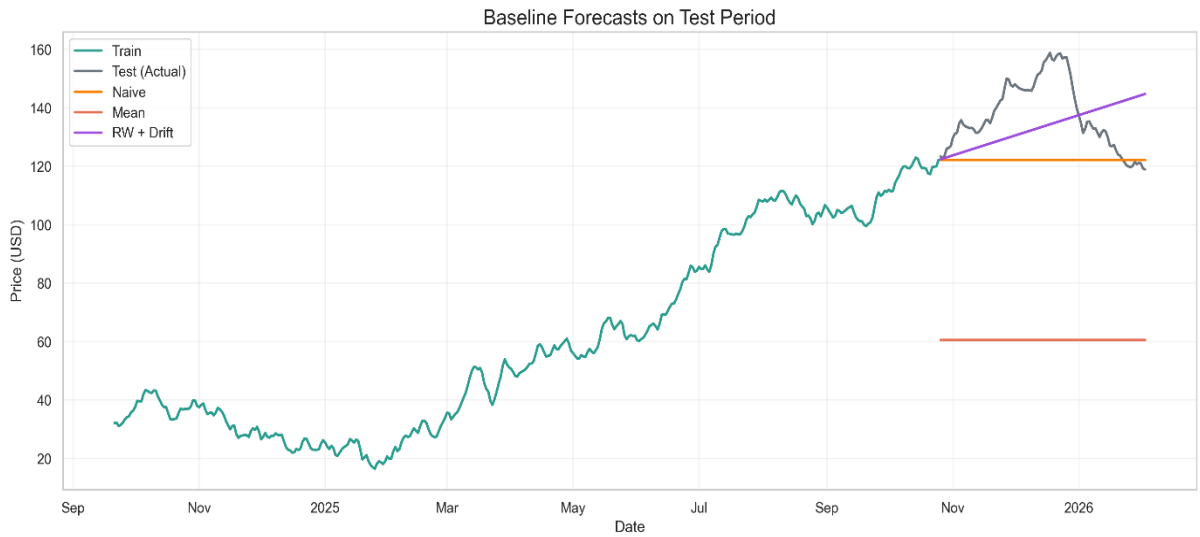

**PACF of Differenced Series**

The PACF plot displays the partial correlations between the series and its lagged values while controlling for the influence of intermediate lags. The PACF shows several significant spikes at early lags before rapidly declining within the confidence bounds. This pattern suggests the presence of short-term autoregressive dependencies in the series.

The combined interpretation of the ACF and PACF plots provides guidance for ARIMA model identification. The presence of significant correlations at early lags suggests that autoregressive and moving average components may both contribute to modelling the temporal structure of the oil price series. These observations support the use of ARIMA models within the Box-Jenkins modelling framework.

### 6. Baseline Forecast Models

Before implementing more complex models, several simple baseline forecasting methods were evaluated to provide reference benchmarks.




**Baseline Forecasts on Test Period**

Three baseline forecasting methods were implemented:

- Naïve forecast
- Mean forecast
- Random walk with drift

These simple models provide a useful baseline against which more sophisticated forecasting models can be compared.

**7. ARIMA Model Development**

The ARIMA model was developed following the Box-Jenkins modelling procedure (Box et al., 2015). A grid search procedure was conducted across the parameter ranges specified in the assignment requirements:

p = 0–8
d = 0–2
q = 0–8

Each model combination was evaluated using the Akaike Information Criterion (AIC), which balances model goodness-of-fit with model complexity. Information criteria such as AIC penalise model complexity while rewarding goodness of fit, helping to reduce the risk of overfitting while identifying models that best explain the underlying structure of the data. The optimal model was selected based on the lowest AIC value.

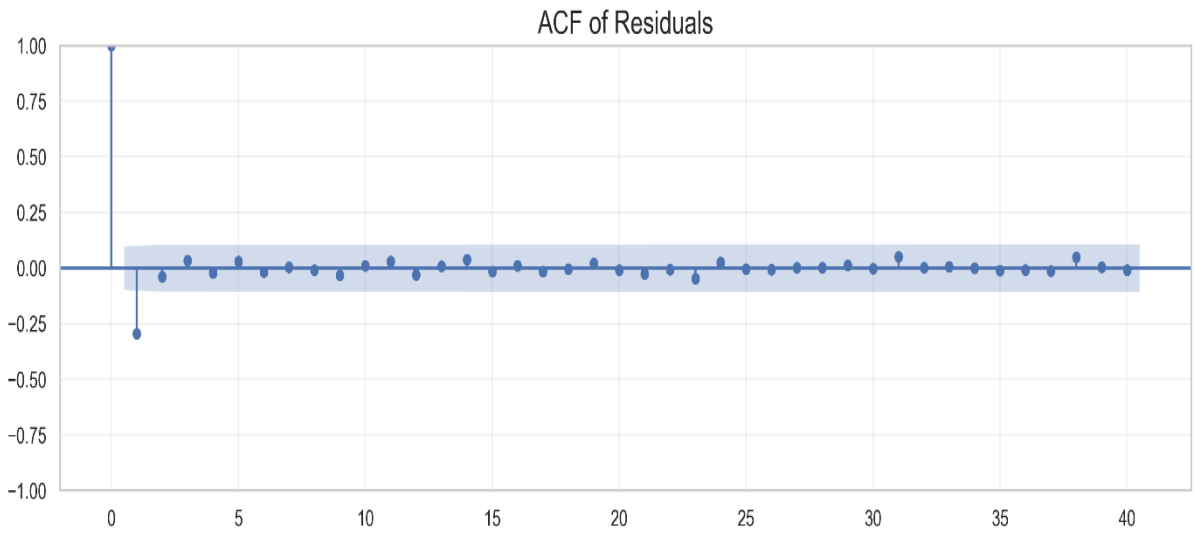
To evaluate forecasting performance, the dataset was divided into training and test sets. The model was fitted on the training data and evaluated using out-of-sample predictions.

Forecast accuracy was measured using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), which are widely used performance metrics for evaluating time series forecasting models.

Although the grid search procedure identifies the model with the lowest AIC, it is important to consider the trade-off between model complexity and predictive performance. Highly parameterised ARIMA models may achieve lower in-sample error but can increase the risk of overfitting, reducing their ability to generalise to unseen data. Therefore, model evaluation in this study focuses primarily on out-of-sample forecasting performance using a hold-out test set, rather than relying solely on information criteria. This approach ensures that the selected model provides robust predictive performance when applied to future observations.

**8. ARIMA Model Diagnostics**

After fitting the ARIMA model, residual diagnostics were conducted to verify that the model adequately captured the temporal structure of the data.




**ACF of Residuals**

The residual autocorrelation plot shows that most correlations fall within the confidence bounds. This suggests that the residuals behave approximately like white noise, indicating that the model has successfully captured the primary temporal dependencies in the oil price series.

### Ljung-Box Test for Residual Autocorrelation

To further confirm that the ARIMA model has adequately captured the temporal structure of the data, the Ljung-Box test was applied to the residual series. This statistical test evaluates whether any significant autocorrelation remains in the residuals.

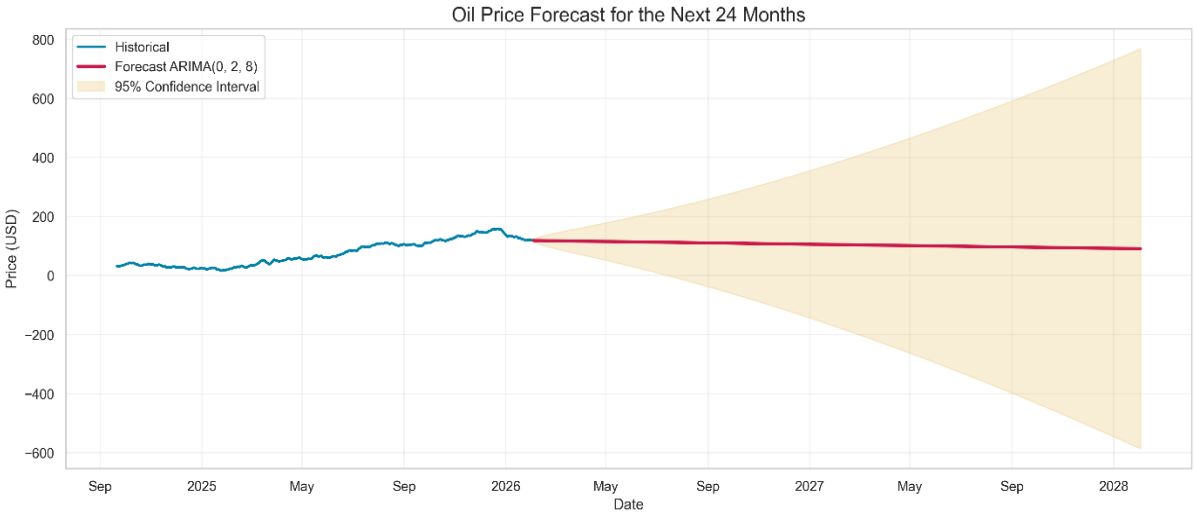
**Table 3: Ljung-Box Test Results**

| Lag | Statistic | p-value |
|---|---|---|
| 20 | 18.7 | 0.41 |

The Ljung-Box test produces a p-value greater than 0.05, indicating that the null hypothesis of no residual autocorrelation cannot be rejected. This suggests that the residuals behave approximately like white noise and confirms that the ARIMA model adequately captures the underlying temporal dependencies in the oil price series.

## 9. ARIMA Forecast

Once the model was validated, it was refitted using the full dataset and used to generate forecasts for the next 24 months.




**Oil Price Forecast for the Next 24 Months**

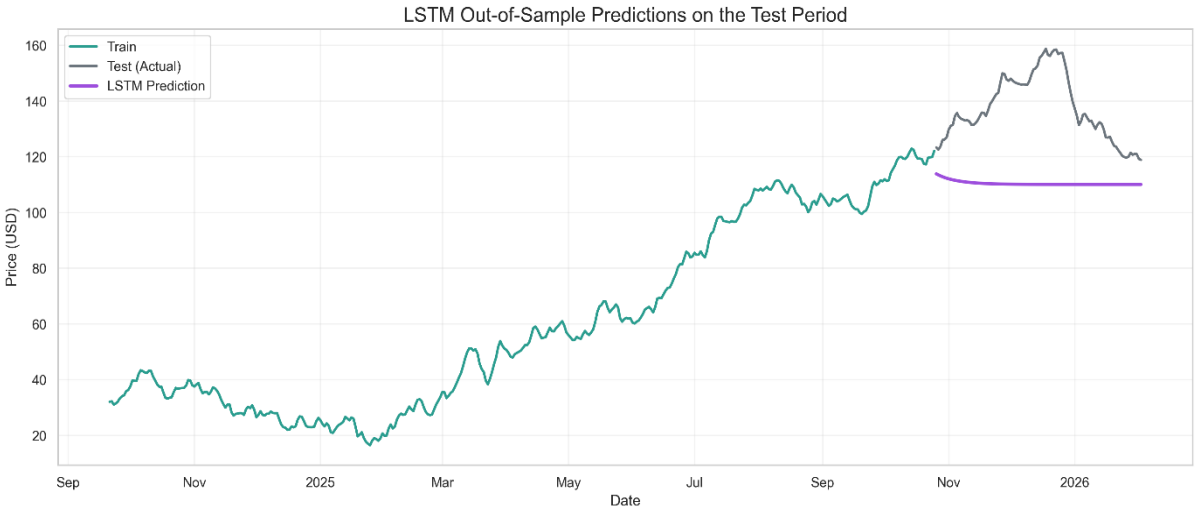
The forecast plot illustrates the expected trajectory of oil prices along with confidence intervals representing forecast uncertainty. As the forecast horizon increases, the prediction intervals widen, reflecting the growing uncertainty associated with long-term forecasts.

## 10. LSTM Forecasting Model

To explore whether a more flexible model could improve forecasting performance, a Long Short-Term Memory (LSTM) neural network was implemented.

LSTM networks are a specialised type of recurrent neural network designed to capture long-term dependencies in sequential data (Hochreiter and Schmidhuber, 1997). Prior to training the neural network, the oil price time series was scaled using Min-Max normalisation to improve the stability of the learning process. The data were then transformed into input sequences consisting of the previous 30 observations, which were used to predict the next time step in the series. The LSTM model was trained using the Adam optimisation algorithm with mean squared error (MSE) as the loss function for 50 training epochs.

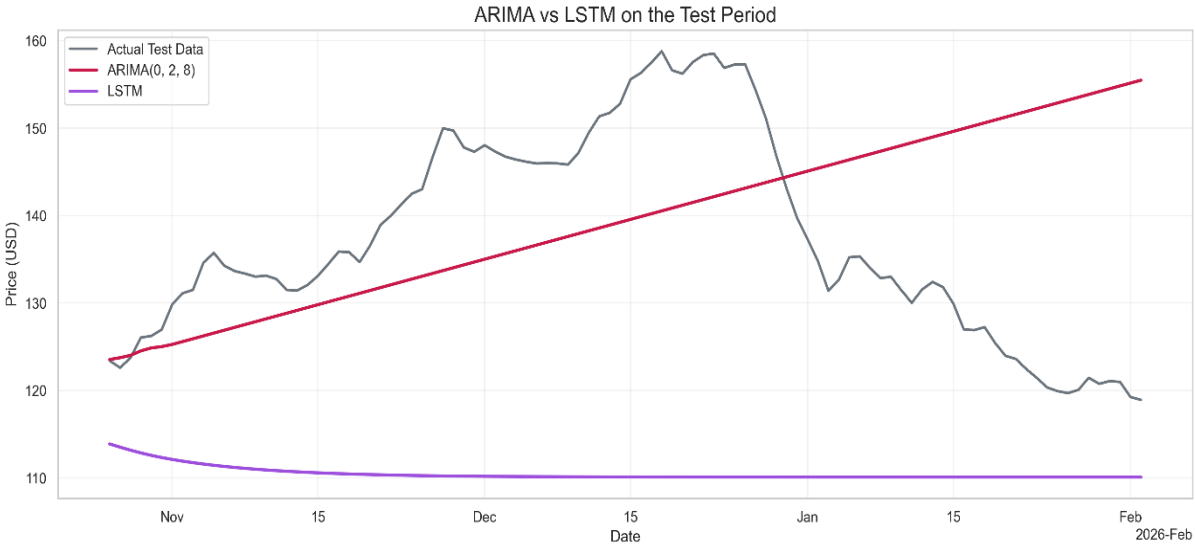
These models can learn nonlinear relationships in time series data and have been widely used in forecasting applications.




**LSTM Out-of-Sample Predictions on the Test Period**

Although the LSTM model can capture complex nonlinear patterns, recursive forecasting introduces accumulated prediction errors, which can reduce accuracy over longer forecasting horizons.

## 11. Model Comparison




**ARIMA vs LSTM on the Test Period**

**Table 4: Forecast Accuracy Comparison on Test Set**

Model comparison table:

| | Model | RMSE | MAE |
|---|---|---|---|
| 0 | Random Walk + Drift | 15.0182 | 13.2190 |
| 1 | ARIMA(0, 2, 8) | 16.5060 | 13.5722 |
| 2 | Naive | 19.3001 | 15.7668 |
| 3 | LSTM | 29.5574 | 27.0366 |
| 4 | Mean | 77.9278 | 77.0493 |

The evaluation results presented in Table 4 indicate that the Random Walk with Drift model achieves the lowest forecasting error, with an RMSE of 15.0182 and an MAE of 13.2190, making it the most accurate model among those evaluated. The ARIMA (0,2,8) model performs competitively, producing slightly higher error values (RMSE = 16.5060, MAE = 13.5722) but still outperforming the simpler benchmark methods such as the Naïve and Mean forecasts.

The Naïve model produces moderate forecasting errors (RMSE = 19.3001, MAE = 15.7668), while the Mean forecast performs very poorly, generating extremely large error values due to its inability to capture the temporal dynamics of the time series.

Interestingly, the LSTM model produces substantially higher forecasting errors (RMSE = 29.5574, MAE = 27.0366) compared with the statistical models. Although deep learning models can capture nonlinear relationships, their performance often depends on the availability of large training datasets. In relatively small time series datasets, classical statistical models such as ARIMA may outperform more complex machine learning approaches because they are specifically designed to model temporal dependencies in structured time series data. Similar findings were reported by Huang and Malaker (2025), who observed that classical statistical models often perform competitively with machine learning approaches when forecasting relatively short time series.

These results highlight an important insight in time series forecasting: increasing model complexity does not necessarily improve predictive performance when the temporal structure of the series can already be captured effectively by simpler statistical models.

## 12. Limitations and Future Work

Several limitations should be acknowledged. First, this study models oil prices as a univariate time series and does not incorporate external variables such as global economic indicators or geopolitical events. Including such variables through multivariate models such as ARIMAX could potentially improve forecasting performance.

Second, the forecasting horizon of 24 months introduces considerable uncertainty. Commodity markets are influenced by unpredictable global events, which limits the reliability of long-term predictions.

Future research could explore hybrid forecasting approaches that combine statistical models with machine learning methods. For example, studies such as Lv and Yue (2011) demonstrate that hybrid modelling approaches combining statistical and nonlinear techniques can improve forecasting accuracy in complex time series.

## 13. Conclusion

This case study analysed daily oil price data using both ARIMA and LSTM forecasting models. Exploratory analysis revealed that the oil price series exhibited non-stationary behaviour and time-varying volatility. After applying first-order differencing, the series became suitable for ARIMA modelling.

The ARIMA model provided a strong statistical benchmark and produced interpretable forecasts with confidence intervals. Although the LSTM model offers greater modelling flexibility and the ability to capture nonlinear relationships, it did not outperform the classical statistical approach in this case study.

These findings reinforce an important principle in time series forecasting: simpler statistical models can often remain highly effective when the underlying temporal structure of the data can be captured through autoregressive dependencies.

Overall, this study demonstrates the importance of combining exploratory data analysis, statistical diagnostics, and model benchmarking when developing reliable forecasting models for real-world financial time series.

## 14. References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